Ideation Phase
Brainstorm & Idea Prioritization Template

Date	31 January 2025
Team ID	LTVIP2026TMIDS80982
Project Name	Prosperity Prognosticator: Machine Learning for Startup Success Prediction
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Reference: https://www.mural.co/templates/brainstorm-and-idea-prioritization

Step-1: Team Gathering, Collaboration and Select the Problem Statement

The team conducted a collaborative discussion session to identify real-world problems in the startup and investment domain. During the discussion, we analyzed the increasing number of startups launching every year and the high failure rate among them.

After evaluating multiple ideas, the team selected the following problem statement:

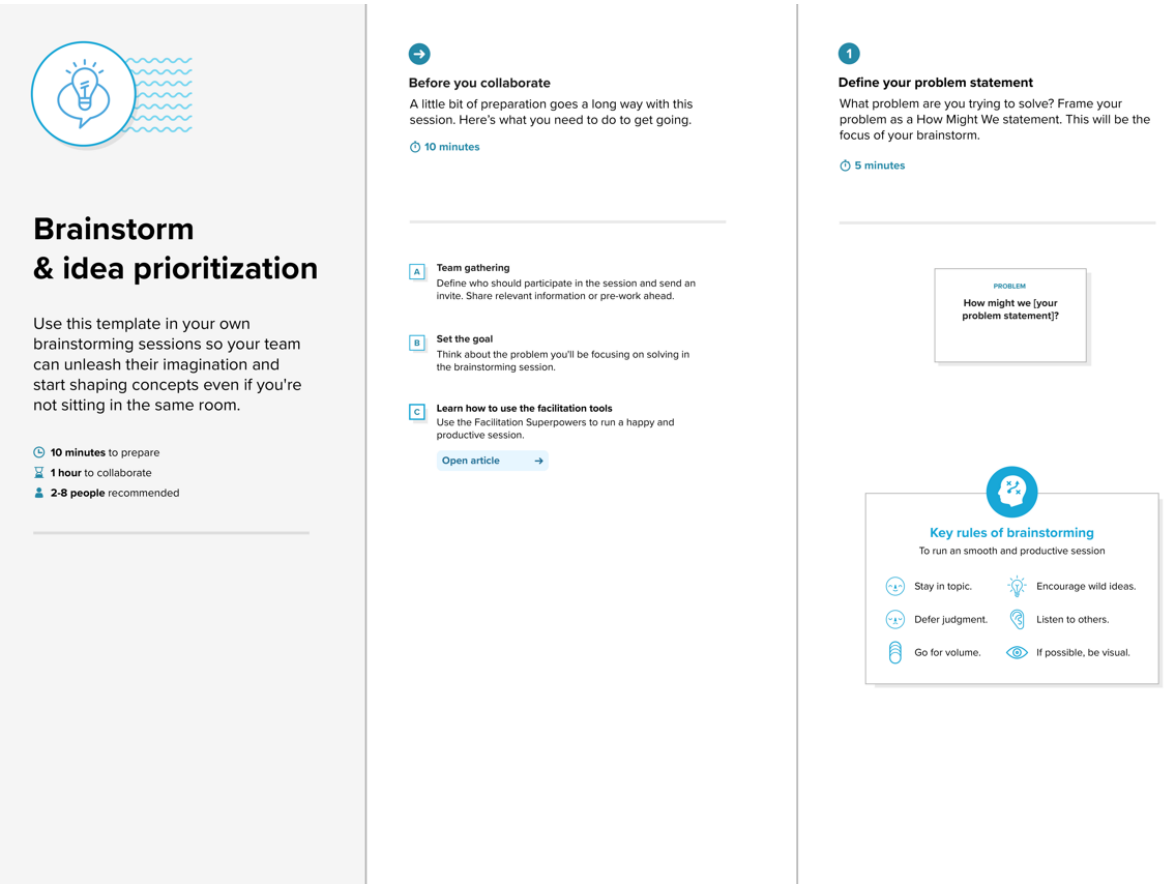
Problem Statement:

To develop a Machine Learning based system that predicts whether a startup is likely to succeed or fail based on funding history, milestones, and relationship data.

The problem was selected because:

- Many startups fail due to poor planning and funding management.
- Investors face difficulty in identifying promising startups.
- Entrepreneurs need data-driven insights before making business decisions.
- There is a growing need for intelligent startup evaluation systems.

All team members agreed that this problem has strong practical value and real-world business impact



Step-2: Brainstorm, Idea Listing and Grouping

During the brainstorming session, team members freely shared ideas without judging or rejecting any concept. The focus was on generating maximum ideas.

Ideas Generated:

- Build a startup success prediction model.
- Use classification algorithms like Logistic Regression and Random Forest.
- Analyze funding rounds and total funding amount.
- Study milestone completion patterns.
- Evaluate relationships and average participants.
- Create a Flask-based web application.
- Display prediction result clearly (Acquired / Closed / Operating).
- Show probability score of success.
- Implement model improvement using retraining.
- Add data visualization for funding analysis.

Idea Grouping:

After listing ideas, we grouped them into categories:

1. Data & Model Ideas

• Use supervised learning classification
• Train models using startup dataset
• Feature selection (funding_total_usd, milestones, relationships)
• Evaluate using Accuracy, Precision, Recall

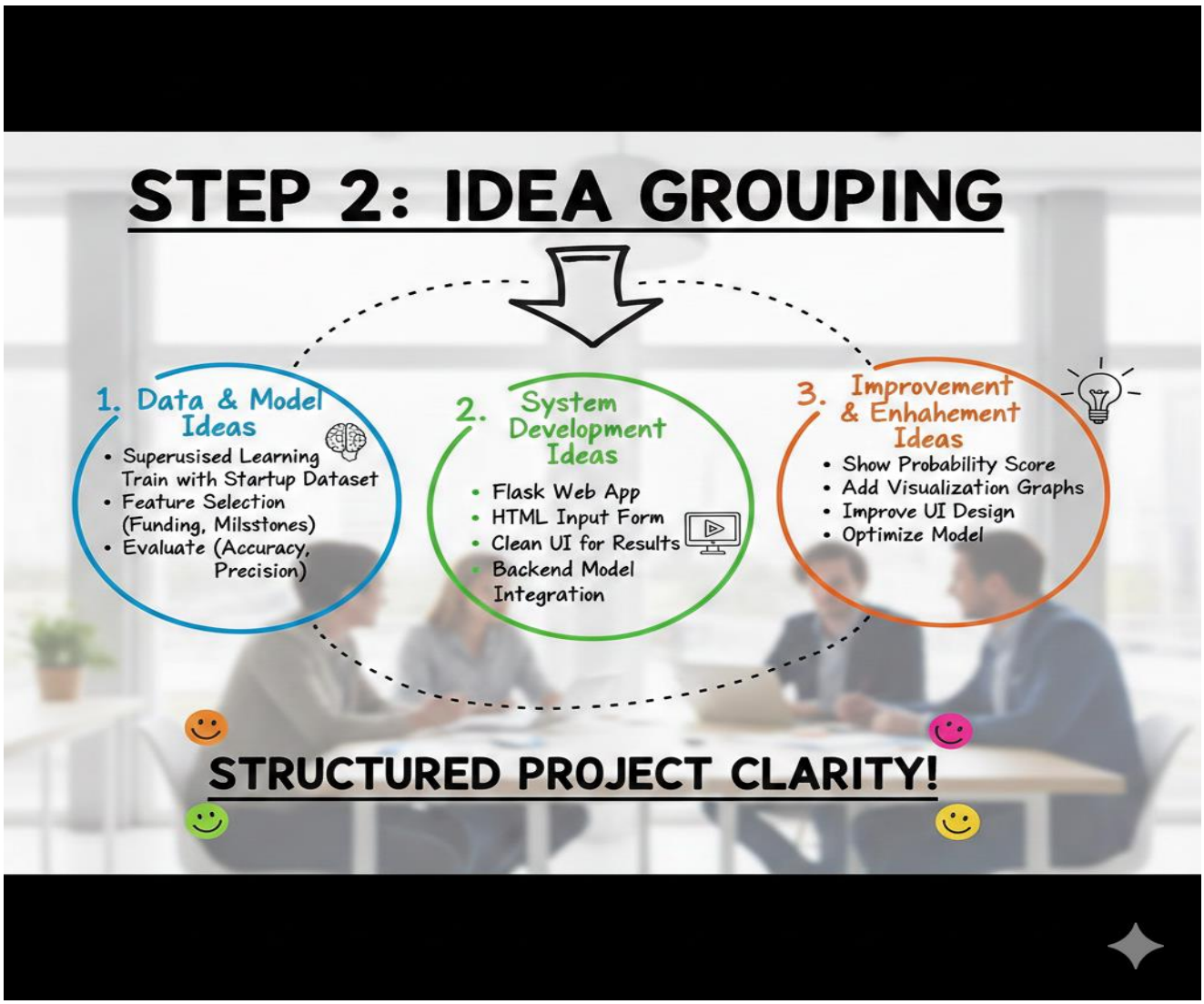
2. System Development Ideas

• Develop Flask web application
• Create HTML input form for startup parameters
• Display prediction result in a clean UI
• Deploy model with backend integration

3. Improvement & Enhancement Ideas

• Show probability score
• Add visualization graphs
• Improve UI design
• Optimize model performance

This grouping helped the team structure the project clearly.



Step-3: Idea Prioritization

- After brainstorming, the team prioritized ideas based on:

  • Feasibility
  • Business Impact
  • Implementation Time
  • Technical Complexity

- **High Priority Ideas:**

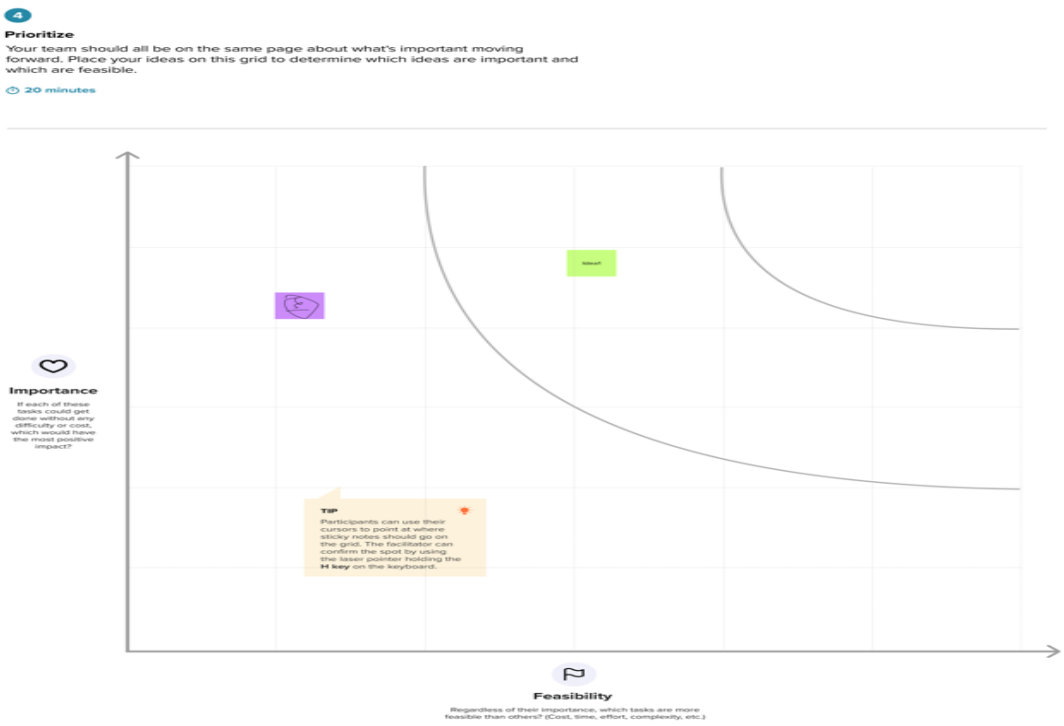
  • Build a classification-based startup prediction model.
  • Use Random Forest for better accuracy.
  • Develop Flask-based web interface.
  • Display clear prediction result (Success / Failure).

- **Medium Priority Ideas:**

  • Add probability score display.
  • Implement model comparison (Logistic vs Random Forest).
  • Add simple data visualization.

- **Low Priority Ideas:**

  • Advanced dashboard analytics.
  • Complex UI animations.
  • Cloud deployment optimization.



Final Selected Approach:

The team decided to:

• Use supervised machine learning classification.
• Train the model on startup funding dataset.
• Select important features like funding rounds, milestones, relationships.
• Evaluate performance using Accuracy, Precision, Recall, and F1-score.
• Deploy the model using Flask with a user-friendly interface.

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